

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO2_AT1 — CHART CONSTRUCTION TEST

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO2 – Construction (BL3)	Assessment:	Assessment Tool 1 – Chart Construction
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO2 Statement:	<i>Apply appropriate chart construction techniques — correct chart type, axes, scales, and calculations — to accurately represent given data.(BL3)</i>		
Student Name:	Tejaswini P	Reg. No.:	192324046
Section / Batch:	2023/B tech AIDS	Date:	30/7/2026

Code of Conduct

I Tejaswini P (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Tejaswini

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 75 minutes.
- Graph paper or a ruler is recommended for accurate, to-scale drawing.
- Show all calculations (e.g., percentages, angles, bin ranges, five-number summaries) as part of your answer.
- Every chart must include a title, correctly labeled axes, and a legend where applicable.

Annexure A – CHART CONSTRUCTION

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT1 — Chart Construction Test

CO2 · BT3: Apply ·

This rubric is used to assess student responses to the CO2-AT1 Chart Construction Test, in which students are given a small dataset and must construct an accurate, correctly labeled chart of a specified type.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of the correct chart type, and the axes, scale, and elements required to construct it for the given data.	Good understanding of the required chart type and construction elements, with only minor gaps.	Basic understanding of chart construction, but with some misconceptions about the correct chart type, axes, or scale.	Poor understanding of core construction concepts; selects a chart type or elements fundamentally unsuited to the data.
Explanation	25%	Shows detailed, clearly worked calculations (e.g., angles, bin ranges, five-number summary) and explains each construction step.	Shows clear calculations and construction steps, with adequate explanation for most parts.	Calculations or construction steps are shown but incomplete, or explanations are superficial.	Little or no working shown; calculations and construction steps are missing or unexplained.
Application	20%	Effectively applies chart construction techniques to produce a complete, accurate chart correctly matched to the data.	Applies construction techniques to produce a correct chart, with only minor errors.	Limited application of construction techniques; the chart partially represents the data but has notable errors.	Fails to apply appropriate construction techniques; the chart does not correctly represent the data.
Organization	15%	The chart is neatly and logically constructed, with a title, correctly labeled axes, a legend where applicable, and elements in a clear order.	The chart is well organized, with only minor issues in labeling, legend, or layout.	Basic construction; required elements (title, axes, legend) are present but poorly arranged or partly missing.	Poorly constructed; the chart is missing key elements such as a title, axis labels, or a legend.
Accuracy	10%	The chart is completely accurate — all values,	Mostly accurate, with only minor omissions or a small error in one	Partially correct; some plotted values, proportions, or	Largely inaccurate; major errors in plotted values,

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		proportions, scales, and calculations correctly match the given data.	value or calculation.	calculations are missing or incorrect.	proportions, or calculations relative to the given data.

Scoring Guide

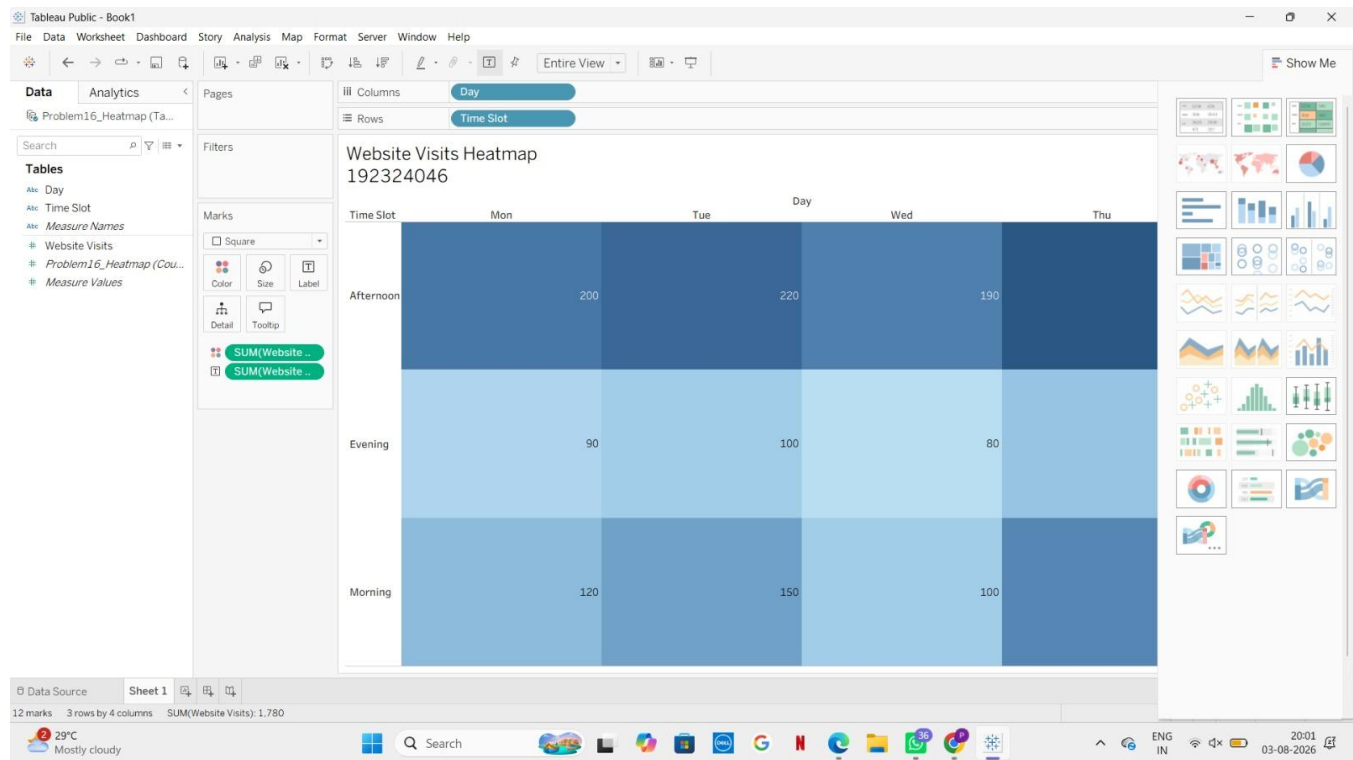
For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem 16 — Heatmap — Grid Construction

Data: Website visits by day and time slot: Morning: Mon=120, Tue=150, Wed=100, Thu=180 Afternoon: Mon=200, Tue=220, Wed=190, Thu=240 Evening: Mon=90, Tue=100, Wed=80, Thu=110

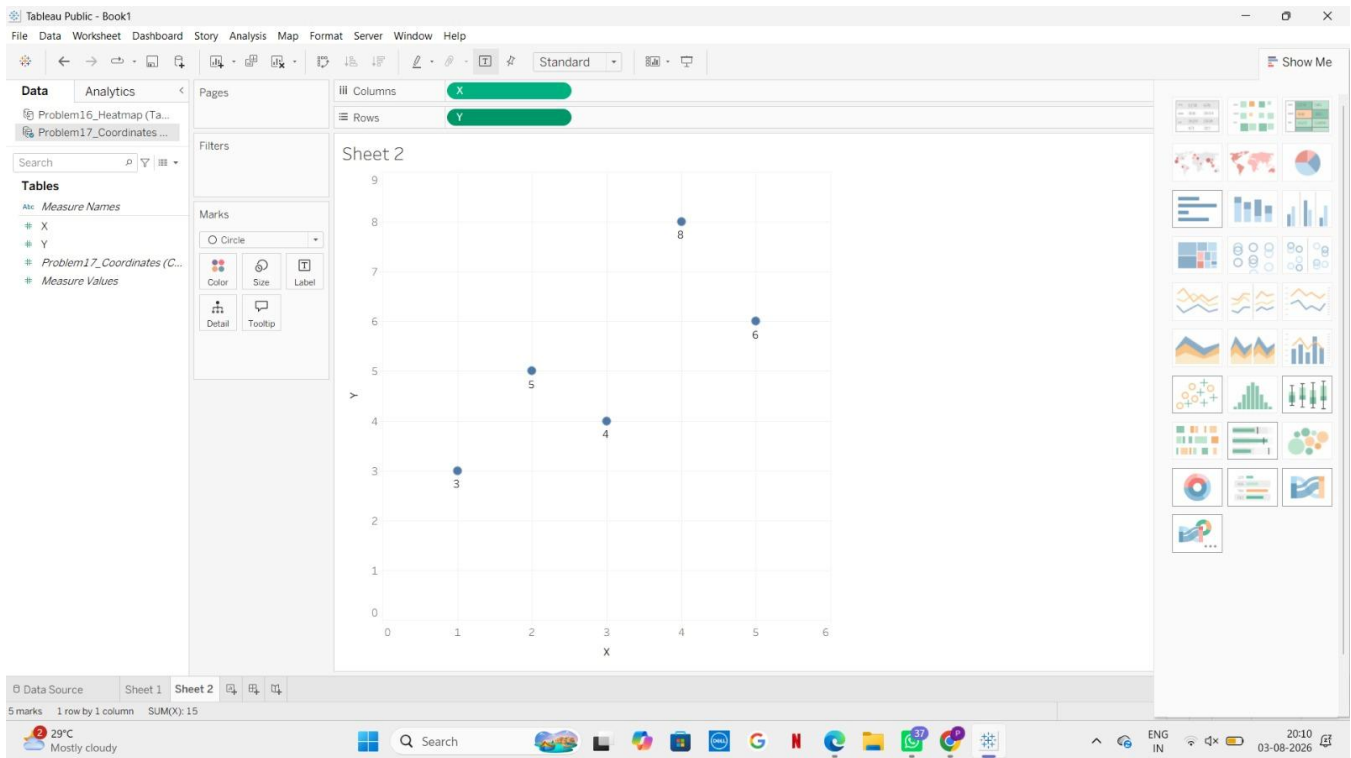
Task: Construct a heatmap for this 3×4 grid, choosing an appropriate sequential color scale and shading each cell according to its value.



Problem 17 — Cartesian Coordinate Plotting

Data: Five (x, y) coordinate points: (1,3), (2,5), (3,4), (4,8), (5,6).

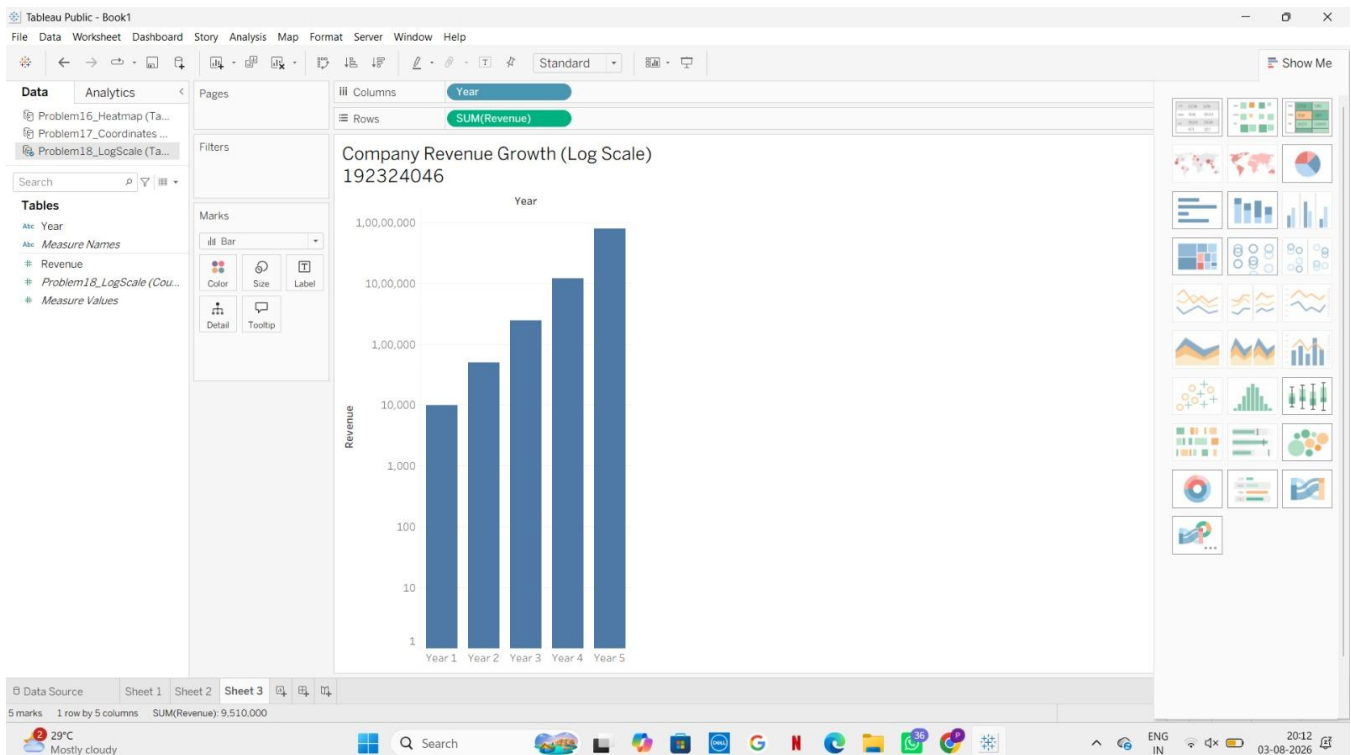
Task: Construct a Cartesian coordinate plot of these 5 points, with correctly scaled and labeled X and Y axes.



Problem 18 — Logarithmic Axis Construction

Data: Company revenue over 5 years: Year 1 = \$10,000; Year 2 = \$50,000; Year 3 = \$250,000; Year 4 = \$1,200,000; Year 5 = \$8,000,000.

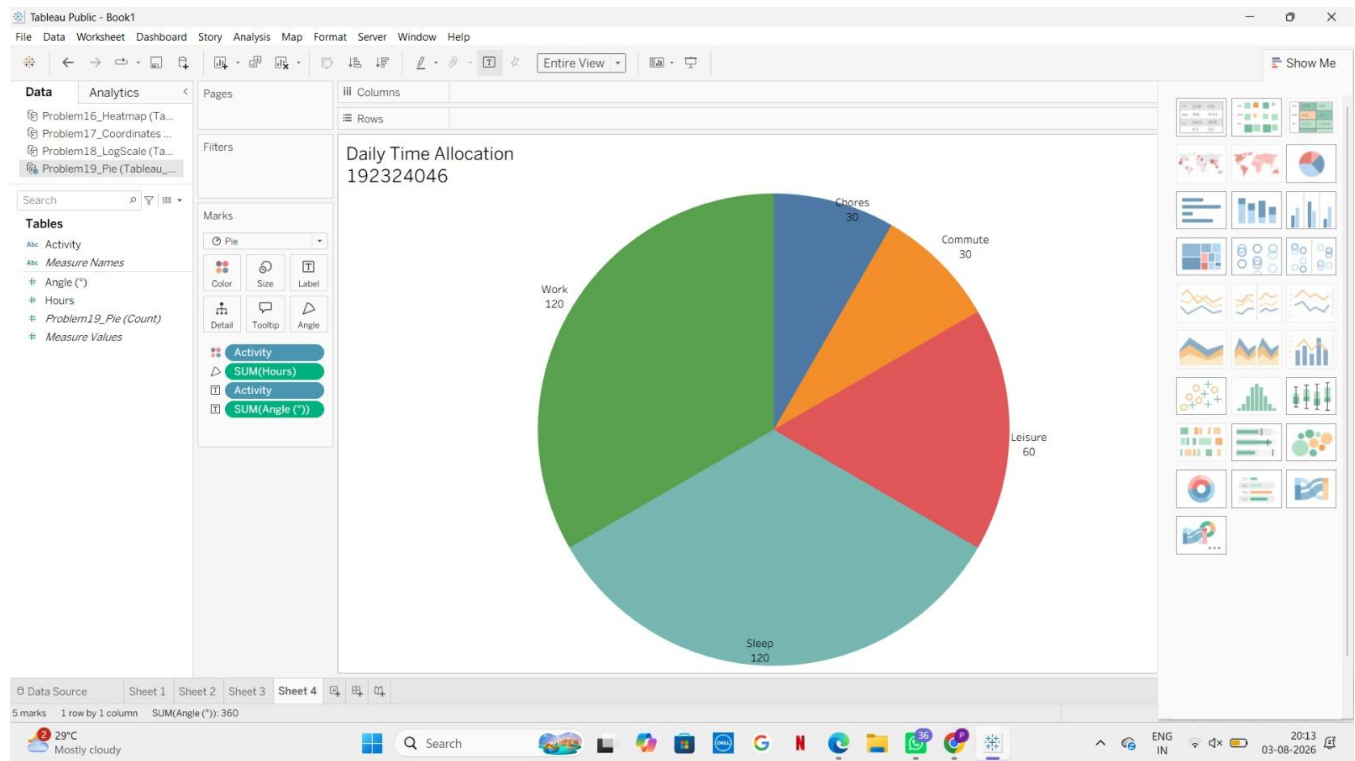
Task: Construct a bar or line chart using a logarithmic y-axis. Show the axis tick values you selected (e.g., powers of 10) and explain why a linear axis would be unsuitable.



Problem 19 — Polar Coordinates — Time Allocation

Data: Time allocation in a 24-hour day: Sleep=8 hrs, Work=8 hrs, Leisure=4 hrs, Chores=2 hrs, Commute=2 hrs.

Task: Construct a pie chart (polar coordinates) showing this time allocation. Calculate and label the angle in degrees for each activity.



Problem 20 — Ordinal Bar Chart — Correct Ordering

Data: Customer satisfaction survey responses: Poor=15, Fair=30, Good=90, Excellent=65.

Task: Construct a bar chart with categories correctly ordered from Poor to Excellent (not alphabetically or by frequency), and briefly justify why this ordering matters for ordinal data.

